Department of Computer Science and Engineering
Winter Semester: 2025-26
CS4086E: System Programming Lab

gcc Practice Questions

Question 1
Write a shell script that:
1. Takes a C file name as an argument.
2. Checks if the input file has a .c extension and exists in the current directory.
3. Compiles the file into an object file (.o), a preprocessed file (.i), and an assembly file (.s).
4. Creates an executable named final_exec.
5. Prints appropriate success/failure messages after each stage.

Question 2
Write a shell script that:
1. Accepts a list of C source files as arguments.
2. Checks if all files exist.
3. Compiles each file into object files (.o).
4. Links all object files into an executable named project_exec.
5. If a source file is modified after the object file is created, recompile only that source file.

Question 3
Write a script that accepts a C file and a compilation stage flag (-E, -S, -c) as inputs, then performs the corresponding stage using gcc.

Question 4
Write a script that:
1. Compiles a given C program through each stage of compilation (.i, .s, .o, and executable).
2. Logs all errors and warnings from each stage into a separate (compile.log) file.
3. Displays the number of warnings and errors after the compilation process.

Question 5
Write a script that compiles a C file (for ex., main.c), that uses a shared library (libarith.so), and links it to generate an executable. The libarith library implements the four arithmetic operations such as addition, subtraction, multiplication, and division.

gdb/valgrind practice question

1. Create a C program tarea.c that calculates the area of a triangle with integer coordinates.
   a) The program should take three sets of (x, y) coordinates, compute the area using the formula: Area=(|x1· (y2−y3)+x2· (y3−y1)+x3· (y1−y2)|)/2
   b) Use gdb to set a breakpoint at the first scanf line where the program reads the first set of coordinates.
   c) Step through the code line by line using next and print the values of x1 and y1 after the first input.

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2. Create a C program fib.c that computes Fibonacci numbers recursively
   a) Set a watchpoint on the variable n to monitor its changes during recursion.
   b) Step through the program and inspect how n changes in each recursive call.

3. Consider the following C -Programming code

```
#include <stdlib.h>
void f(void) {
    int* x = malloc(10 * sizeof(int));
    x[10] = 0;
}
int main(void) {
    f();
    return 0;
}
```

   Debug the program using valgrind memory check.

4. Consider the following program :

```
#include <stdio.h>
    int main() {
    int t;
    int result = 10 / t;
    printf("Result: %d\n", result);
    return 0;
}
```

   a) Use valgrind to identify the issue and then debug the program with gdb to understand the root cause.
   b) Fix the code and verify using both tools.

5. The following code contains two functions: one that inadvertently leaks memory and one that uses a pointer after it has been freed. Use GDB and Valgrind to identify these issues and correct the code.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void fn_leak() {
    char *str = malloc(50);
    if (!str) return;
    strcpy(str, "This is an english sentence to be printed using C program.");
    printf("fn_leak: %s\n", str);

    return;
}
```

```
void fn_dangle() {
   int *ptr = malloc(sizeof(int));
   if (!ptr) return;
   *ptr = 42;
   free(ptr);

   printf("Pointer value: %d\n", *ptr);
}

int main() {
   fn_leak();
   fn_dangle();
   return 0;
}
```

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